An Energy-Based View on the Manifold Hypothesis

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Abstract

A commonly adopted assumption in modern machine learning is the *manifold hypothesis*, which claims that although data is embedded in a high-dimensional ambient space, the probability mass concentrates near a much lower-dimensional geometric structure. However, the real-world latent variables that images are drawn from are continuous, not discrete, and therefore cannot be completely captured by a finite manifold. Moreover, the assumption does not properly capture the union of manifolds, stratified spaces, and manifolds with singularities and boundaries. Drawing upon recent advances in energy-based models, we provide an energy-based perspective on the real-world latents, describing an Occam's razor style analog for the manifold hypothesis, which is more amenable to recent literature in the field. We theoretically prove that our construction recovers the manifold hypothesis for coarse functions, and validate it with numerical examples.

1 Introduction

If pictures are discretized objects in $\mathbb{R}^d$, where $d = n^2$ is the dimension of the image.

We consider raw objects $f : [0,1]^2 \to \mathbb{R}^3$ corresponding to depixelated RGB pictures.

A commonly adopted assumption in modern machine learning is the *manifold hypothesis*, which claims that although data is embedded in a high-dimensional ambient space, the probability mass concentrates near a much lower-dimensional geometric structure.

Assumption 1.1 (Manifold Hypothesis)**.** Let $\chi \subset \mathbb{R}^D$ be the ambient space and let $\mathcal{D}$ be a probability distribution on χ. There exists a d-dimensional manifold $\mathcal{M} \subset \chi$, with $d \ll D$, such that

$$\mathbb{P}_{x \sim \mathcal{D}}\big(\mathrm{dist}(x, \mathcal{M}) \leq \varepsilon\big) \geq 1 - \delta,$$

for sufficiently small $\varepsilon, \delta > 0$.

In practice, data rarely concentrate on a single smooth manifold. Rather, they are better modeled as lying on *unions of manifolds*, or more generally on *stratified spaces* composed of pieces of varying intrinsic dimensions, potentially with singularities and boundaries.

Why small d is important?

- Generalization error depends on the intrinsic dimension d of the manifold.
- The number of samples n needed to learn the manifold increases with d.
- The complexity of the model increases with d.
- Can get a small embedding.

Some shortcomings of the manifold hypothesis are that

- Doesn't properly capture union of manifolds.

- Doesn't properly capture stratified spaces.
- Doesn't properly capture manifolds with singularities and boundaries.
- What about infinite-dimensional data? Cause images come from infinite-dimensional continuous functions.

[Emile says: would be nice to elaborate on the above, and to point to specific instances where assuming it has some "catastrophic" consequences. that could go into a motivating examples section]

1.1 Related Works

Energy-based models Song and Kingma [2021], Du and Mordatch [2019], LeCun et al. [2006], Zhu et al. [2024], Du et al. [2021, 2020]

Compositional generation Su et al. [2024], Liu et al. [2021, 2022, 2023], Du and Kaelbling [2024], Bagautdinov et al. [2018]

Occam's razor Elmoznino et al. [2025], Rasmussen and Ghahramani [2000]

Example: Vision Data Each 2d image (or 3d scene or a point cloud) can be modeled $f : \mathbb{R}^2 \to \mathbb{R}^3$ as a function from coordinates to RGB (or XYZ) values. That's how NeRFS work:

Neural Radiance Fields (NeRFs). Mildenhall et al. [2021], Pumarola et al. [2021], Xu et al. [2021], Wang et al. [2022], Rabby and Zhang [2024], Reiser et al. [2021], Müller et al. [2022], Tancik et al. [2021], Yu et al. [2021], Chen et al. [2022] [Emile says: this actually looks very relevant. would be worth doing a deeper dive into this] A *Neural Radiance Field* models a 3D scene as a continuous function

$$f_\theta : (\mathbf{x}, \mathbf{d}) \;\mapsto\; (\sigma, \mathbf{c}),$$

where

- $\mathbf{x} \in \mathbb{R}^3$ denotes a spatial location,
- $\mathbf{d} \in \mathbb{S}^2$ denotes a viewing direction,
- $\sigma \in \mathbb{R}_{\geq 0}$ is the volume density at $\mathbf{x}$,
- $\mathbf{c} \in \mathbb{R}^3$ is the emitted RGB color in direction $\mathbf{d}$.

The key idea is that the scene is represented neither as a mesh, nor as a voxel grid, nor as a point cloud, but as a *continuous radiance field* parameterized by a neural network.

In a similar way, each 2d image can be modeled as a function from coordinates to RGB values: $f : \mathbb{R}^2 \to \mathbb{R}^3$ and the embedding of this function would the weights of the corresponding ReLU network. The regular manifold hypothesis doesn't properly capture this structure that:

- In a union of manifolds
- Has some hierarchical structure.
- Has permutation-invariance with respect to weights in each layer.

Also, importantly, each image f is not a smooth function: its a combination of a "simple" in some sense partition of the image into smooth enough regions.

2 Set Up

We work in an abstract setting that captures the common statistical belief that observed data $x_1, \ldots, x_n \in X$ concentrate on a low-dimensional structure, even when the ambient space X is high-dimensional. Concretely, we fix a *reference* distribution ν_0 on X (e.g. Gaussian or uniform) encoding our prior knowledge about X, and posit that the data are drawn from a *target* distribution μ that is far more concentrated — e.g. supported on a d-dimensional manifold $M \subset \mathbb{R}^D$ with $d \ll D$. The benefit is quantitative: greater concentration of μ yields faster learning rates for functions $f : X \to \mathbb{R}$.

Differential structure. To differentiate functions on X we fix an *algebra of test functions* $\mathcal{A}$ (a subset of $\mathbb{R}^X$ closed under pointwise products) and a *derivation*

$$D : \mathcal{A} \longrightarrow \mathcal{M}$$

into a module $\mathcal{M}$ equipped with a pointwise inner product $\langle \cdot, \cdot \rangle_x$, satisfying the Leibniz rule $D(fg) = fDg + gDf$. This gives rise to the *carré du champ* operator

$$\Gamma(f, g)(x) := \langle Df(x),\, Dg(x) \rangle_x,$$

which generalizes the square-gradient $|\nabla f|^2$ to the abstract setting.

Dirichlet form and Laplacian. Given any σ-finite measure ν on $(X, \mathcal{F})$, the carré du champ integrates to a *Dirichlet form*:

Definition 2.1 (Symmetric Dirichlet Space [Bakry et al., 2014, §3.4])**.** Let $(X, \mathcal{F}, \mu)$ be a σ-finite measure space. A *Dirichlet form* on $L^2(\mu)$ is a pair $(\mathcal{E}, \mathcal{D}(\mathcal{E}))$ where:

1. $\mathcal{D}(\mathcal{E}) \subset L^2(\mu)$ is a dense linear subspace;
2. $\mathcal{E} : \mathcal{D}(\mathcal{E}) \times \mathcal{D}(\mathcal{E}) \to \mathbb{R}$ is a symmetric bilinear form;
3. **(Non-negativity)** $\mathcal{E}(f, f) \geq 0$ for all $f \in \mathcal{D}(\mathcal{E})$;
4. **(Closedness)** $\mathcal{D}(\mathcal{E})$ is complete under $\|f\|_{\mathcal{E}}^2 := \|f\|_{L^2(\mu)}^2 + \mathcal{E}(f, f)$;
5. **(Markov property)** for every $f \in \mathcal{D}(\mathcal{E})$, the truncation $\widetilde{f} := (0 \vee f) \wedge 1$ satisfies $\widetilde{f} \in \mathcal{D}(\mathcal{E})$ and $\mathcal{E}(\widetilde{f}, \widetilde{f}) \leq \mathcal{E}(f, f)$.

The quadruple $(X, \mathcal{F}, \mu, \mathcal{E})$ is called a *Symmetric Dirichlet Space*.

In our setting the Dirichlet form is

$$\mathcal{E}_\nu(f, g) = \int_X \Gamma(f, g)\, d\nu,$$

so $\mathcal{E}_\nu(f, f)$ is the total gradient energy of f under ν. The Markov property is automatic from the Leibniz rule and the truncation behaviour of Γ.

Definition 2.2 (Generalized Laplacian)**.** To any Dirichlet form $(\mathcal{E}_\nu, \mathcal{D}(\mathcal{E}_\nu))$ on $L^2(\nu)$ there corresponds a unique self-adjoint operator L_ν on $L^2(\nu)$ satisfying

$$\mathcal{E}_\nu(f, g) = \langle f,\, L_\nu g \rangle_{L^2(\nu)}, \qquad \forall\, f \in \mathcal{D}(\mathcal{E}_\nu),\; g \in \mathcal{D}(L_\nu).$$

L_ν admits a spectral decomposition $L_\nu \phi_j = \lambda_j \phi_j$ with

$$0 = \lambda_0 \leq \lambda_1 \leq \lambda_2 \leq \cdots, \qquad \langle \phi_j, \phi_k \rangle_{L^2(\nu)} = \delta_{jk}.$$

The Markov property ensures that e^{-tL_ν} is a Markov semigroup: starting from any initial distribution ν_0 on X, the heat flow $e^{-tL_\nu}\nu_0$ converges to ν as $t \to \infty$, confirming that ν is the invariant measure of the diffusion. The *iterated carré du champ* Γ_2^ν measures how Γ evolves under this semigroup:

$$\Gamma_2^\nu(f, g) = \tfrac{1}{2}\left(L_\nu \Gamma(f, g) - \Gamma(f, L_\nu g) - \Gamma(g, L_\nu f)\right), \qquad \Gamma_2^\nu(f) := \Gamma_2^\nu(f, f).$$

The Bakry–Émery condition $\Gamma_2^\nu(f) \geq \rho\, \Gamma(f)$ ($\rho \in \mathbb{R}$) encodes a lower bound on the Ricci curvature of the underlying space.

Spectral dimension. The eigenvalue asymptotics of L_ν encode the intrinsic dimension of the learning space, providing the key rate parameter throughout the paper.

Definition 2.3 (Effective Dimension)**.** The *effective dimension* of $(X, \mathcal{F}, \nu, \mathcal{E}_\nu)$ is the quantity $d_{\text{eff}} \in (0, \infty)$ characterised by any of the following equivalent asymptotics:

$$\begin{aligned} \lambda_j &\asymp j^{2/d_{\text{eff}}} && \text{(eigenvalue growth),} \\ N(\lambda) := \#\{j : \lambda_j \leq \lambda\} &\asymp \lambda^{d_{\text{eff}}/2} && \text{(counting function),} \\ \operatorname{tr} e^{-tL_\nu} &\sim t^{-d_{\text{eff}}/2} && \text{(heat-kernel trace, as } t \to 0^+\text{).} \end{aligned}$$

For a compact d-dimensional Riemannian manifold M, Weyl's law gives $d_{\text{eff}} = d$, confirming that the effective dimension recovers the geometric dimension in the classical setting.

3 Effective Dimension Hypothesis

Assumption 3.1 (Effective Dimension Hypothesis)**.** All the observed data $x_1, \ldots, x_n \in X$ come from a learning space $(X, \mathcal{A}, \mu)$ with small effective dimension even if the ambient space $(X, \mathcal{A}, \nu_0)$ has large effective dimension where ν_0 is the base distribution on X:

$$d_{\text{eff}}(\mu) \ll d_{\text{eff}}(\nu_0)$$

4 Relevant Things

Definition 4.1 (Curvature–Dimension Condition, Ledoux [2000] (Definition 1.2))**.** For $\rho \in \mathbb{R}, N \in (0, \infty)$, L_ν satisfies $\mathrm{CD}(\rho, N)$ if for all $f \in \mathcal{A}$

$$\Gamma_2^\nu(f, f) \geq \rho\Gamma(f, f) + \frac{1}{N}\left(L_\nu f\right)^2$$

Intuitively: $\rho \sim$ curvature parameter, $N \sim$ dimension

Example 4.2. *Examples of generators L_ν that satisfy the CD condition:*

- *An n-dimensional complete Riemannian manifold (M, g) with Ricci curvature bounded below, or rather the Laplacian Δ on M, satisfies the inequality $\mathrm{CD}(R, n)$ with R the infimum of the Ricci tensor and n the geometric dimension.*
- *The classical Laplacian Δ on $\mathbb{R}^n$ satisfies $\mathrm{CD}(0, n)$,*
- *the Laplace-Beltrami operator on the unit sphere S^n is of curvature dimension $\mathrm{CD}(n-1, n)$ since curvature is constant and equal to $n-1$ in this case.*

4.1 Sub-Exponential Concentration

Definition 4.3 (Log-Sobolev Inequality)**.** Given μ for there is a constant C_{LS} such that the following holds for any $f : X \to \mathbb{R}$

$$\operatorname{Ent}_\mu\left(f^2\right) \leq 2C_{LS} \int \|Df\|^2 d\mu$$

where $\operatorname{Ent}_\mu(g) = \int g \log g d\mu - \int g d\mu \cdot \log \int g d\mu$ for $g \geq 0$.

Lemma 4.4. *Assuming $\Gamma_2^\mu(f, f) \geq \rho\Gamma(f, f)$ for all $f \in \mathcal{A}$ then $C_{LS} \leq \frac{1}{\rho}$*

Consequences:

Theorem 4.5 (Hypercontractivity, Gross 1975)**.** *The LSI is equivalent to hypercontractivity of the semigroup: for $1 < p < q < \infty$,*

$$\|P_t f\|_{L^q(\mu)} \leq \|f\|_{L^p(\mu)} \quad \textit{when } t \geq \frac{C_{LS}}{2} \log \frac{q-1}{p-1}$$

Theorem 4.6 (Sub-Gaussian concentration)**.** *For f with $\Gamma(f, f) \leq 1$,μ–a.e., the Herbst argument gives*

$$\mu\left(f - \mathbb{E}_\mu f \geq t\right) \leq \exp\left(-\frac{t^2}{2C_{LS}}\right)$$

The smaller C_{LS} – the better.

4.2 Effective Dimension

Corollary 4.7. *If ν satisfies* $\mathrm{CD}(\rho, N)$ *then* $d_{\mathrm{eff}} \leq N$

5 Supervised Learning: Learning a Function From Data

Adding a prior $p(f) \propto e^{-E(f)}$ over functions $f : X \to \mathbb{R}$ the supervised learning problem is equivalent to the following optimization problem:

$$\widehat{f}_n = \min_f \sum_{i=1}^{n} \|f(x_i) - y_i\|^2 + E(f)$$

Note, that $E(f)$ defines sublevel sets $\mathcal{F}_r = \{f : E(f) \leq r\}$. The generalization error of $\widehat{f}_n$ decomposes as:

$$\mathbb{E}\left\|f^* - \widehat{f}_n\right\|_{L^2}^2 \leq \underbrace{\inf_{f : E(f) \leq r} \|f - f^*\|^2}_{\text{bias: how well } \mathcal{F}_r \text{ approximates } f^*} + \underbrace{\varepsilon_n^2(r)}_{\text{variance: estimation error within } \mathcal{F}_r}$$

Definition 5.1 (Sublevel set)**.** Given a convex functional $E : \mathcal{A} \to \mathbb{R} \cup \{+\infty\}$, a radius $r > 0$, and a metric d on $\mathcal{A}$, the sublevel set is:

$$\mathcal{B}_r(E) = \{f \in \mathcal{A} : E(f) \leq r\}$$

Definition 5.2 (Covering number)**.** The ϵ-covering number is:

$$N(\epsilon, \mathcal{B}_r(E), d) = \min\left\{m \in \mathbb{N} : \exists f_1, \ldots, f_m \in \mathcal{A} \text{ s.t. } \mathcal{B}_r(E) \subseteq \bigcup_{i=1}^{m} \{g : d(g, f_i) \leq \epsilon\}\right\}$$

The natural metric in our setting is $d(f, g) = \|f - g\|_{L^2(\mu)} = \left(\int_X (f - g)^2 d\mu\right)^{1/2}$

Definition 5.3 (Metric entropy)**.**

$$\mathcal{H}(\epsilon, \mathcal{B}_r(E), d) = \log N(\epsilon, \mathcal{B}_r(E), d)$$

The smaller the metric entropy – the less samples we need to learn a function from data.

Lets consider 3 simple enough cases of what $E(f)$ can be:

1. Given convex $g : \mathbb{R} \to \mathbb{R}$ define $E(f) = \int_X g(f(x)) d\mu(x)$
2. $E(f) = \mathcal{E}_\mu(f, f)$
3. Given $\phi : \mathbb{R} \to \mathbb{R}$ define $E(f) = \langle f, \phi(L_\mu) f\rangle$

Proposition 5.4. *For $E(f) = \int_X g(f(x)) d\mu(x)$ The covering number of $\mathcal{B}_r(E)$ is at most $N(r, E) = \frac{1}{r}\int_0^r \mathcal{H}(\epsilon, \mathcal{B}_r(E), d) d\epsilon$.*

As we introduced Definition 4.1 that captures the "compactness" of the measure, we can say now that $\mu \ll \nu$ if

$$\mu \in CD(\rho_\mu, N_\mu), \nu \in CD(\rho_\nu, N_\nu)$$
$$\text{and}$$
$$\rho_\mu > \rho_\nu, N_\mu < N_\nu$$

The following theorem captures what density $\frac{d\mu}{d\nu}$ "simplifies" the measure.

Theorem 5.5 (Perturbation of CD(ρ, N))**.** *[TODO: double check, add a proof, better notation]*

Let $(X, \mathcal{A}, \Gamma, \mu_0)$ be a Markov diffusion structure with generator L_0 satisfying CD(ρ, N) :

$$\Gamma_2^{(0)}(f, f) \geq \rho\Gamma(f, f) + \frac{1}{N}(L_0 f)^2 \quad \textit{for all } f \in \mathcal{A}$$

Let $U \in \mathcal{A}$ satisfy:

- *Abstract κ-convexity:* $\Gamma(f, \Gamma(U, f)) - \frac{1}{2}\Gamma(U, \Gamma(f, f)) \geq \kappa\Gamma(f, f)$ *for all f*
- *Bounded abstract gradient:* $\Gamma(U, U) \leq M$ *μ_0-a.e.*

Let $d\mu = Z^{-1}e^{-U}d\mu_0$ with $L_U f = L_0 f - \Gamma(U, f)$. Then for any $\varepsilon \in (0, 1)$, (L_U, Γ, μ) satisfies $\mathrm{CD}\,(\rho', N')$ *with*

$$\rho' = \rho + \kappa - \frac{(1/\varepsilon - 1)M}{N}, \quad N' = \frac{N}{1 - \varepsilon}$$

In particular, $C_{LS}(\mu) \leq \frac{1}{\rho'}$ whenever $\rho' > 0$.

6 Diffusion Models

[TODO: Add noise schedule maybe.]

Forward process: $\mu \to \mu_0$ Density $p_t = \mu_t/\mu_0$ evolves according to the diffusion equation:

$$\frac{\partial p_t}{\partial t} = L_{\mu_0} p_t \implies p_t = e^{tL_{\mu_0}} \mu_0$$

Backward process: $\mu_0 \to \mu$ Density p_t evolves according to the diffusion+drift equation:

$$\frac{\partial p_t}{\partial t} = L_{\mu_0} p_t - \nabla \cdot (p_t \nabla \log p_t)$$

To be able to reverse it back, we need to learn $\nabla \log p_t$. Let us look at the very first δ step of the forward process (corresponding to the very last δ step of the backward process) and learn a score function for that step $s(\cdot) = \nabla \log p_T = \nabla \log \frac{d\mu}{d\nu}$. The minimization problem is:

$$\min_s \underbrace{\mathbb{E}_{p_T}\left[\|s(\cdot) - \nabla \log p_T\|^2\right]}_{\text{data term}} + \underbrace{R(s)}_{\text{regularizer term}} \tag{1}$$

While the data term is straightforward, the regularizer term is to be chosen. Assuming $d\mu = Z^{-1}e^{-U}d\mu_0 \Rightarrow \nabla \log p_T = -DU$. Our prior (hope) for U is that it concentrates the measure better, specifically, that by Definition 4.1 $U \in \mathcal{A}$ satisfies:

- κ-convexity: $\Gamma(f, \Gamma(U, f)) - \frac{1}{2}\Gamma(U, \Gamma(f, f)) \geq \kappa\Gamma(f, f)$ for all f
- Bounded gradient: $\Gamma(U, U) \leq M$ μ_0-a.e.

Define the abstract Hessian as a map from 1-forms to bilinear forms on $D(\mathcal{A})$:

$$H_v(f) := \langle Df, D\langle v, Df\rangle\rangle - \frac{1}{2}\langle v, D\langle Df, Df\rangle\rangle$$

This is linear in v : $H_{\alpha v_1 + \beta v_2}(f) = \alpha H_{v_1}(f) + \beta H_{v_2}(f)$. Note, that the term $\langle v, v\rangle \leq M$ is quadratic in v.

Hence, the following set is convex:

$$\mathcal{C}_{\kappa,M} = \{v \in D(\mathcal{A}) : H_v(f) \geq \kappa\Gamma(f, f) \text{ and } \langle v, v\rangle \leq M \quad \forall f \in \mathcal{A}\}$$

First let us compute its covering number.

Lemma 6.1. *The covering number of $\mathcal{C}_{\kappa,M}$ is at most $N(M, \kappa) = \frac{M}{\kappa}$.*

Lemma 6.2 (Very rough sketch)**.** *Define $W_{\kappa,M}$ as [TODO: a fixed set of very cool points from $\mathcal{C}_{\kappa,M}$]. Then for every $v \in \mathcal{C}_{\kappa,M}$ the following problem has a unique solution:*

$$\min_{\omega \in \mathfrak{M}_+(W_{\kappa,M})} \|\omega\|_{TV} \text{ such that } \int_{w \in W_{\kappa,M}} w d\omega(w) = f$$

Hence, if we know the set $W_{\kappa,M}$ then Equation (1) can be rewritten as:

$$\min_{\omega \in \mathfrak{M}_+(W)} \mathbb{E}_{p_T}\left[\left\|\int_{w \in W_{\kappa,M}} w d\omega(w) - \nabla \log p_T\right\|^2\right] + \|\omega\|_{TV}$$

6.1 References

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7 Energy-Based Model Perspective on Effective Dimension

need framework that shows that

- pure noise
- fractals
- matryoshka

should be hard to learn

8 Examples

9 Conclusion

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